ECE496 Proposal

ASAP: Automated Social-media Feed Aggregation Platform

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Executive Summary

The Middleware Systems Research Group (MSRG) at the University of Toronto currently relies on the third-party service Elfsight to aggregate social-media content for its website. The existing workflow provides limited platform coverage and control over content processing. However, posts from LinkedIn take at least 48 hours to be delivered to the MSRG website. ASAP (Automated Social-media Feed Aggregation Platform) is an open-source system designed to provide MSRG with a more efficient and manageable method for aggregating and publishing relevant social-media content.

ASAP is aimed to retrieve posts from supported platforms, including X, Bluesky, and LinkedIn. Provider-specific adapters will access and download posts according to platform requirements and regulations. Retrieved posts will retain their original content and provider-specific presentation information, while a separate normalized representation will be generated for filtering and duplicate detection. This allows equivalent posts across multiple platforms to be identified without modifying the original content ultimately displayed to website visitors.

The platform will also provide an administrator-controlled publishing workflow. New or updated content can be placed in a review queue and communicated to the feed owner through notifications. Through a manageable backend, administrators will be able to review potential duplicates, approve or reject posts, select target websites, choose a preferred source version, reorder or pin content, configure expiration times, remove published items, and create manual announcements. Approved content will be presented through a responsive HTML iframe while preserving recognizable provider-specific styling.

ASAP is designed around a modular architecture consisting of ingestion and monitoring, content processing and persistence, curation, publishing, and presentation subsystems. To ensure reproductivity, the project will provide Docker deployment support along with published documentation. The project would also be open-source on Github to allow for future improvements and development. Preliminary testing has already demonstrated successful retrieval of text, links, and images from X and Bluesky. LinkedIn integration's availability is still dependent on platform authorization; further verification will be conducted once the team obtains an account with business verification. The designated final version of ASAP should be an MSRG-controlled social-media aggregation platform that reduces dependence on third-party services, improving the timeliness and manageability of website updates, and provides a maintainable foundation for future providers and websites.

Table of Contents

1.0 Introduction.....1

2.0 Background and Motivation..... 1

3.0 Problem Statement.....2

4.0 Project Goal..... 2

5.0 Requirements Specification..... 3

6.0 System Block Diagram..... 5

7.0 Risk Assessment..... 6

8.0 Conclusion..... 7

9.0 References.....8

Appendix A: Project Gantt Chart.....9

Appendix B: Timeline and Documentation..... 10

Appendix C: Preliminary Research.....11

Appendix D: System Block Diagram (3 subsystems).....12

1.0 Introduction

The Middleware Systems Research Group (MSRG) currently relies on a third-party service, Elfsight, to retrieve social-media content for its website. The current provider supports limited platforms and can introduce an update delay of approximately 48 hours. This project proposes **ASAP (Automated Social-media feed Aggregation Platform)**, an open-source platform that provides a more convenient and cost-effective solution to the current problems. ASAP will retrieve content from X, Bluesky, and LinkedIn where access permits, and provide a common workflow for normalization, filtering, duplicate detection, storage, and administrator review. A primary objective is to make relevant content available for administrator review within at least 48 hours after publication, while maintaining a modular architecture that can accommodate future providers with new platform APIs.

2.0 Background and Motivation

Social-media aggregation tools allow organizations to collect content from multiple platforms and present it through a unified website feed. Existing commercial solutions such as Elfsight and Walls.io provide embeddable social-media feeds, while open-source projects such as Harken demonstrate alternative approaches to social-media collection and management [1], [2], [3]. These solutions reduce the effort required to develop individual platform integrations, but differ in platform coverage, customization, backend manageability, and intended use.

The MSRG website (msrg.org) presents the University of Toronto's Middleware Systems Research Group and its research in data and distributed systems, distributed machine learning, and quantum computing systems. It introduces the group's work and provides information for visitors interested in research, collaboration, and graduate or postdoctoral opportunities [4]. The homepage includes a News section, which is the intended website destination for ASAP's curated social-media feed. ASAP will supply this section with relevant posts from supported platforms, allowing visitors to view group updates in one place while giving MSRG control over which content appears.

MSRG currently relies on Elfsight's aggregation service for its website. Although convenient, the current workflow provides limited control over content retrieval, processing, and publication. Moreover, posts from LinkedIn usually have a **48 hours** delay to appear on the website; the current free service is limited to approximately **500 pulls per month**, and the feed primarily supports LinkedIn services only. Alternative commercial services offer similar aggregation capabilities but would still keep much of the

processing workflow outside MSRG's direct control. ASAP instead targets an MSRG-controlled workflow with configurable sources, administrator curation, and website publication.

Preliminary research and feasibility testing were conducted to evaluate both the need for and technical viability of the proposed system. Direct content retrieval was successfully demonstrated using test accounts for **X and Bluesky**, including post text, links, and images. LinkedIn integration was also investigated, but access remains dependent on business-account authentication and platform authorization. These findings indicate that direct multi-platform aggregation is technically feasible while highlighting the importance of isolating provider-specific access constraints from the common processing workflow. Detailed feasibility results and a comparison with existing solutions are provided in **Appendix C**.

ASAP is motivated by MSRG's need for a more timely, manageable, and maintainable social-media workflow. Its primary stakeholders are **MSRG administrators and website maintainers**, who gain greater control and visibility over content aggregation and publication, and **website visitors**, who benefit from more timely updates. The proposed system is intended to combine multi-platform retrieval, normalization, filtering, duplicate detection, administrator review, and publication within a modular platform that can adapt to future provider and website changes.

3.0 Problem Statement

MSRG currently lacks a directly controlled and timely method for aggregating social-media content from multiple platforms for publication on its website. The existing third-party solution introduces significant update delays, limited platform coverage, and limited control over content processing and publication. The engineering problem is to develop a reliable aggregation system that can handle heterogeneous social-media APIs, process content into a consistent form, identify relevant and duplicate posts, and support administrator-controlled publication while remaining maintainable as provider APIs and website requirements evolve.

4.0 Project Goal

The goal of ASAP is to provide MSRG with a timely, manageable, and maintainable system for aggregating and publishing relevant social-media content on its website.

5.0 Requirements Specification

The System Requirements Specification (SyRS) defines the top-level functional and quality requirements, constraints, and objectives for ASAP. Functional and quality requirements define the capabilities and performance necessary for project success, while constraints capture external limitations and objectives describe desirable improvements. These specifications will guide system design and evaluation of the final implementation.

ID	Specification	Description and verification
R-01	ASAP shall retrieve posts from configured LinkedIn, X, and Bluesky sources for which access is available under C-01.	Requirement: Verify retrieval from each accessible provider using authorized test sources; record any inaccessible provider and the reason.
R-02	ASAP shall represent retrieved posts in a common feed format that preserves the source platform, available post content, and link to the original post.	Requirement: Compare representative source posts with their normalized feed entries.
R-03	ASAP shall include or exclude posts according to configured filtering and curation rules.	Requirement: Verify that a test set produces the expected included and excluded posts for each configured rule.
R-04	ASAP shall expose the curated feed through an HTML iframe that can be embedded in a website.	Requirement: Embed the iframe in a test page and verify that the curated feed is displayed.
R-05	ASAP shall permit an additional provider to supply posts to the common feed through a defined integration interface without changing existing provider integrations or common feed processing.	Requirement: Add a test provider and verify its feed output and the continued operation of existing integrations.
R-06	ASAP shall include a repeatable test harness for provider integrations, filtering rules, feed rendering, and iframe embedding.	Requirement: Run the harness and inspect its reported results for all four areas; distinguish fixture-based tests from live-provider tests.
R-07	ASAP shall include documentation for deployment, configuration, maintenance, and addition of a provider integration.	Requirement: Review coverage of all four topics and verify the deployment and extension instructions by following them.

ID	Specification	Description and verification
R-08	ASAP shall provide the curated social feed used in production on msrg.org in place of the current third-party feed.	Requirement: Verify that the production website displays the ASAP feed and obtain MSRG confirmation that it replaces the current feed.
C-01	Provider access shall remain within each platform's permitted access mechanisms, authorization requirements, and usage limits [5], [6], [7].	Constraint: Review the access method and applicable permissions and limits for each provider. Team confirmation required: accessible source and post types for LinkedIn, X, and Bluesky.
C-02	The ASAP project shall be published as open source under a permissive license in the MSRG GitHub organization.	Constraint: Verify repository availability, organization ownership, and the license file. Team confirmation required: license selection.
C-03	ASAP shall support deployment in a Docker container.	Constraint: Deploy and run the feed service using the supplied Docker deployment instructions.
O-01	Minimize recurring operating cost, with free operation preferred.	Objective: Compare estimated monthly hosting and provider-access costs for the agreed usage profile. Team confirmation required: usage profile and acceptable cost ceiling.
O-02	Reduce feed refresh delay, with a target of less than 48 hours.	Objective: Measure elapsed time from a post becoming retrievable through an authorized provider interface to its appearance in the embedded feed. Team confirmation required: test conditions and whether the target becomes a mandatory requirement.
O-03	Enable future use of the ASAP feed on qscc.ca without changes to common feed processing.	Objective: Demonstrate the same feed service embedded in a second test website; deployment on qscc.ca remains a later-stage objective.

6.0 System Block Diagram

Figure 1 presents the high-level architecture of ASAP. The system is divided into three main subsystems: **Ingestion and Monitoring**, **Content Processing and Persistence**, and **Curation, Publishing, and Presentation**.

The **Ingestion and Monitoring subsystem** retrieves content from LinkedIn, X, and Bluesky through provider-specific adapters. A configurable scheduler initiates feed updates at different frequencies for each provider or supported website. The adapters manage OAuth authentication, token renewal, API rate limits, retries, and provider-specific response formats. After retrieving content, the subsystem detects newly created posts and executes download, later explained in subsystem 2. Operational information—including unsuccessful requests, authentication failures, and API rate-limit conditions—is recorded for monitoring and troubleshooting.

The **Content Processing and Persistence subsystem** preserves the original content and metadata obtained from each provider. A separate normalized representation is then derived only for comparison and duplicate detection. For example, the system may normalize text formatting, remove URL tracking parameters, identify common links, and generate media or content signatures. Exact and similarity-based comparisons are used to detect equivalent posts published on the same or different platforms. Duplicate records are associated with a common duplicate group, while their original provider records remain unchanged. The normalized representation is therefore not displayed to users; the published feed uses the preserved original content and provider-specific presentation information.

The processed information is stored in an SQLite database. In addition to original posts and comparison signatures, the database maintains duplicate relationships, per-site publication status, site configuration, manual announcements, source-update history, and administrative audit records. Publication status distinguishes between pending, approved, rejected, published, removed, and expired content.

The **Curation, Publishing, and Presentation subsystem** allows the feed owner to control which content appears on each website. When a new post or a significant source update is detected, the system can send an email notification to the owner and place the content in a review queue. Through the management dashboard, the owner can preview the post, review possible duplicates, approve or reject publication, select the target websites, choose the preferred source version, change the display order, pin important posts, set an expiration time, or remove previously approved content. The dashboard also supports manually created announcements that do not originate from a social media provider.

Finally, the feed service retrieves only approved and non-expired content for the requested website. Provider-specific renderers use the preserved source information to produce recognizable post styles or official platform embeds where appropriate. The resulting feed is delivered through a responsive HTML iframe that supports desktop, tablet, and mobile layouts in major browsers. This design provides a common publishing pipeline while allowing MSRG, QSCC, and future websites to maintain independent feed configurations.

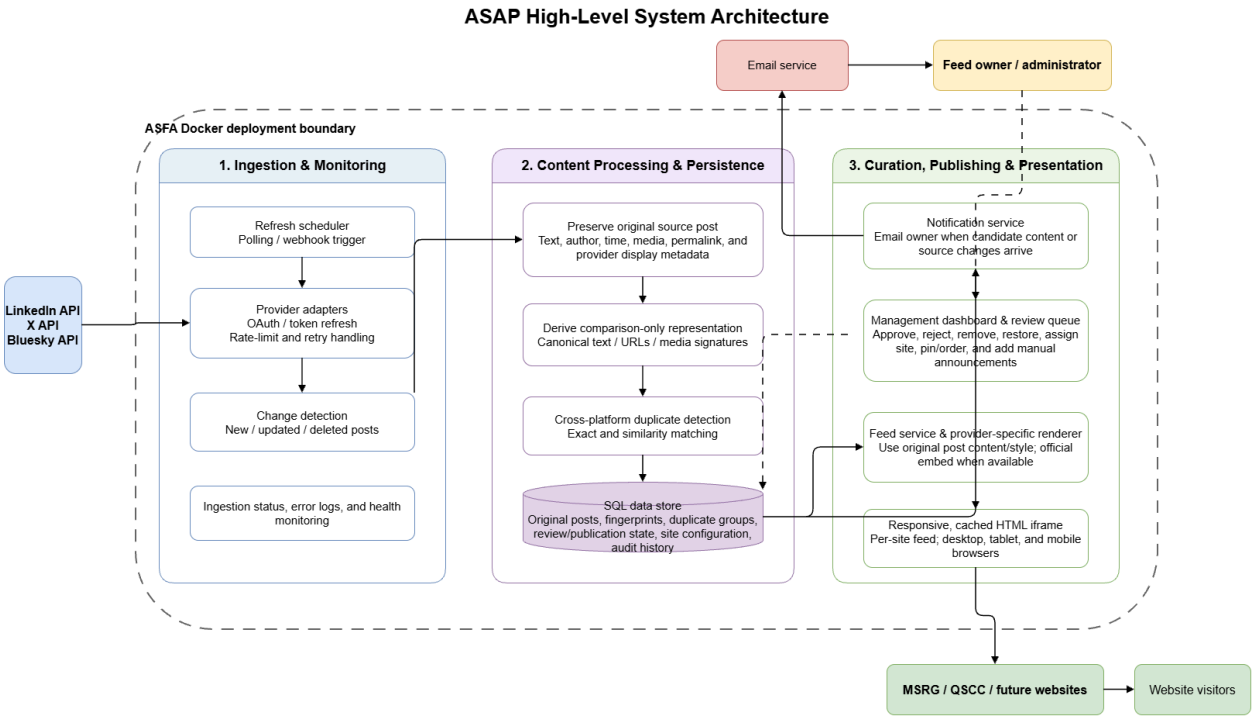


Figure 1. High-level system block diagram of ASAP. (Detailed subsystem in Appendix D)

7.0 Risk Assessment

ASAP interacts with external social-media APIs and provides an administrator-facing management backend, creating potential security risks involving credentials, unauthorized access, and untrusted external content. The primary cybersecurity risks and mitigations are summarized below.

Possible Risk	Level	Consequence	Resolution
Hardcoded API keys or plaintext token storage in source repositories or logs.	Medium	Unauthorized API access, account suspension, and	Store secrets in environment variables, scan code for exposed

Possible Risk	Level	Consequence	Resolution
		unexpected provider quota exhaustion or charges.	credentials, restrict access, and establish secret rotation procedures.
Flaws in backend authentication logic, weak password management, or broken session handling.	Medium	Unauthorized admin access allowing malicious publication, deletion, or tampering with social feed items.	Implement standard authentication libraries, enforce secure session cookies, roll out multi-factor authentication, and conduct access control reviews.
Insufficient sanitization of incoming external social-media content (text, embedded links, HTML snippets).	High	Cross-Site Scripting (XSS) or rendering of malicious links on the official MSRG site.	Treat external data as untrusted, apply robust strict HTML escaping/sanitization libraries, and restrict unsafe inline script execution via Content Security Policy (CSP).
Lack of strict validation schemas for incoming provider payload structures.	Medium	Unhandled parsing exceptions leading to buffer overflows, application crashes, or system denial of service.	Enforce runtime schema validation at adapter ingress, and wrap parsing calls in defensive error-handling blocks.
Missing API rate limiting on public backend endpoints.	Medium	API abuse, automated scraping overhead, or backend server exhaustion.	Implement middleware-based API rate limiting to restrict request frequency and prevent excessive traffic on exposed endpoints.

8.0 Conclusion

ASAP aims to provide MSRG with a timely, manageable, and maintainable system for aggregating and publishing relevant social-media content on its website. The proposed system combines multi-platform content retrieval, data normalization, filtering and deduplication, persistent storage, administrator review, and website feed integration within a modular architecture. By replacing the current third-party aggregation workflow with a group-controlled system, ASAP targets a reduction in content-update latency from approximately **48 hours to within 1 hour** for newly retrievable relevant content.

Preliminary API testing with X and Bluesky supports the technical feasibility of the proposed architecture, while the modular design limits dependencies on individual providers such as LinkedIn. The

project is feasible within the ECE496 development period because the core provider interfaces and processing workflow have already been validated through preliminary prototypes, while remaining external dependencies are isolated from the core system. The resulting platform is expected to give MSRG greater control over its social-media workflow, substantially reduce content-update latency, and provide a maintainable foundation for future providers and website requirements.

9.0 References

[1] Elfsight, “Social Media Feed widget for website.” Accessed: Oct. 9, 2026. [Online]. Available: https://elfsight.com/social-feed-widget/

[2] Walls.io, “The social media aggregator for brands and marketers.” Accessed: Oct. 9, 2026. [Online]. Available: https://walls.io/features/social-media-aggregator

[3] VladUZH, “Harken.” Accessed: Oct. 9, 2026. [Online]. Available: https://github.com/VladUZH/harken

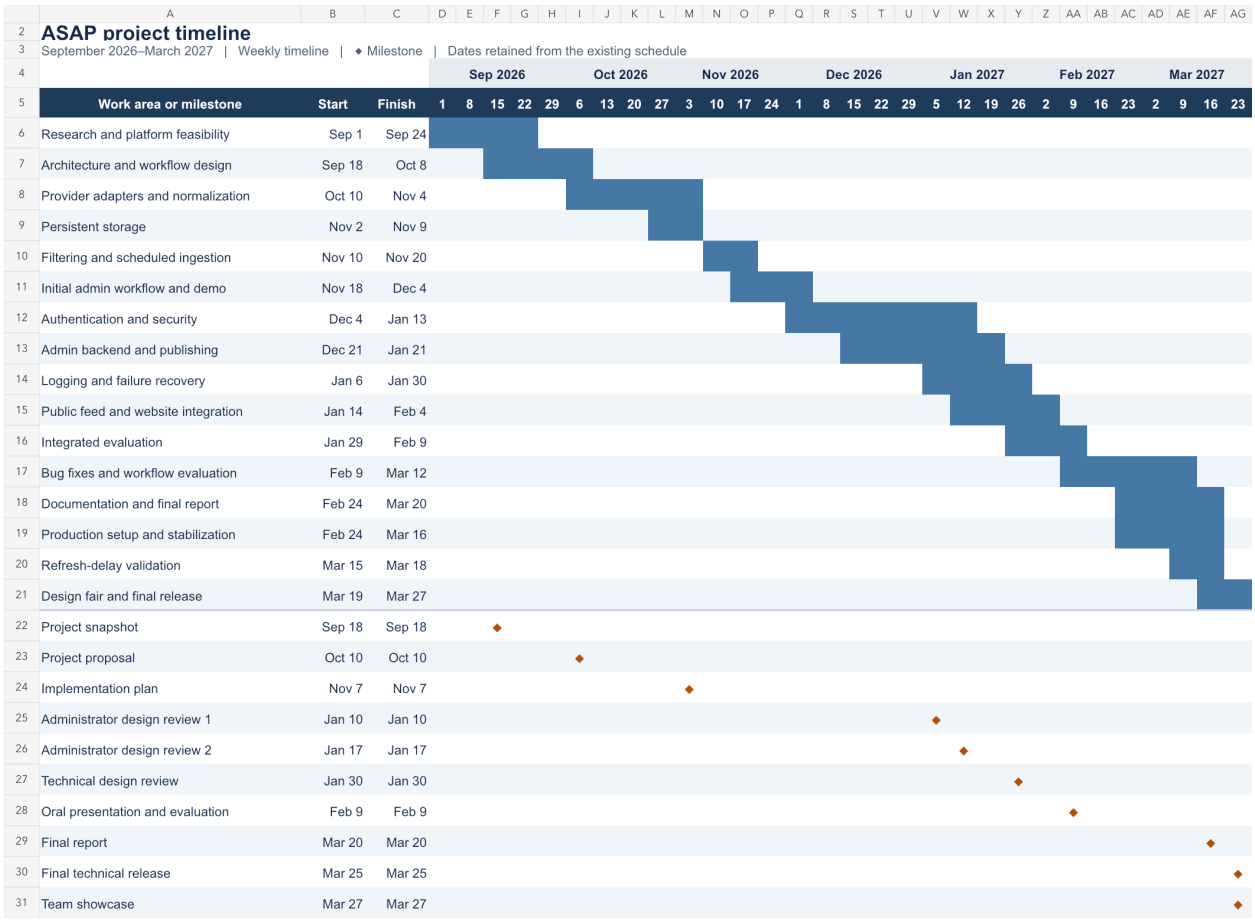
[4] Middleware Systems Research Group, “Home.” Accessed: Oct. 9, 2026. [Online]. Available: https://msrg.org/

[5] X, “X Developer Platform.” Accessed: Oct. 9, 2026. [Online]. Available: https://docs.x.com/overview

[6] Bluesky, “Bluesky HTTP API Reference.” Accessed: Oct. 9, 2026. [Online]. Available: https://endpoints.bsky.app/

[7] LinkedIn, “LinkedIn API Documentation.” Accessed: Oct. 9, 2026. [Online]. Available: https://learn.microsoft.com/en-us/linkedin/

Appendix A: Project Gantt Chart



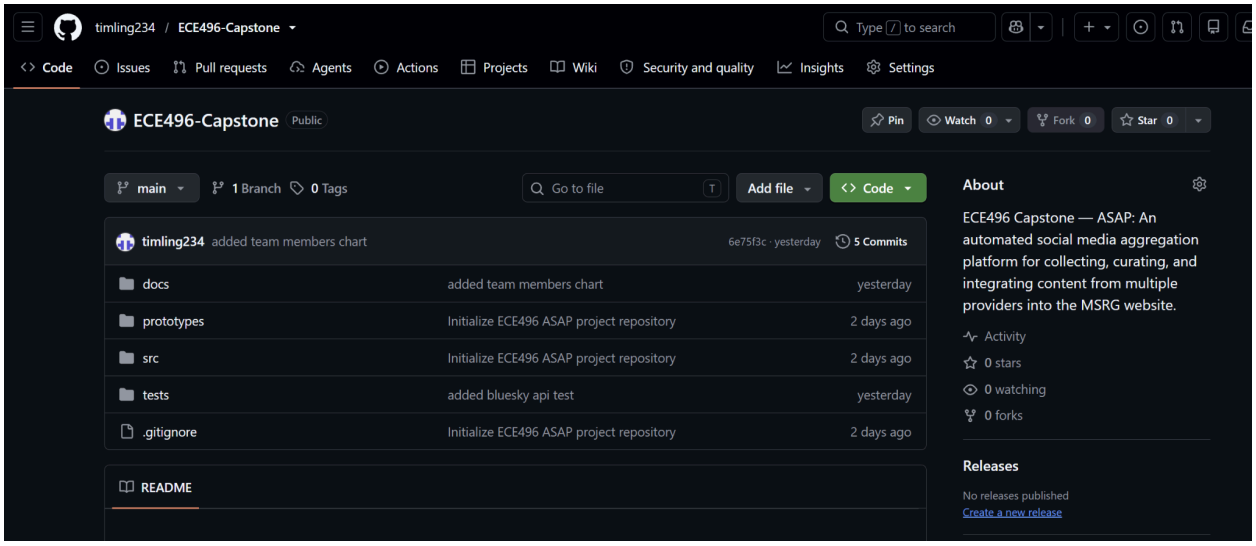
ASAP project timeline

September 2026–March 2027 | Weekly timeline | ◆ Milestone | Dates retained from the existing schedule

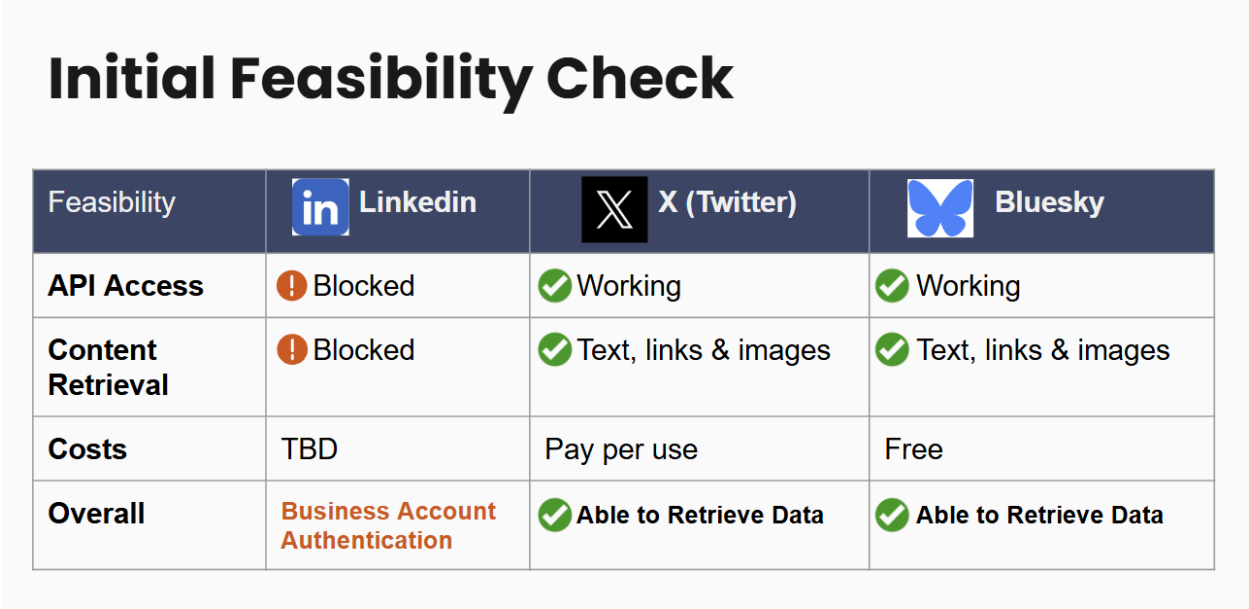
Work area or milestone	Start	Finish
Research and platform feasibility	Sep 1	Sep 24
Architecture and workflow design	Sep 18	Oct 8
Provider adapters and normalization	Oct 10	Nov 4
Persistent storage	Nov 2	Nov 9
Filtering and scheduled ingestion	Nov 10	Nov 20
Initial admin workflow and demo	Nov 18	Dec 4
Authentication and security	Dec 4	Jan 13
Admin backend and publishing	Dec 21	Jan 21
Logging and failure recovery	Jan 6	Jan 30
Public feed and website integration	Jan 14	Feb 4
Integrated evaluation	Jan 29	Feb 9
Bug fixes and workflow evaluation	Feb 9	Mar 12
Documentation and final report	Feb 24	Mar 20
Production setup and stabilization	Feb 24	Mar 16
Refresh-delay validation	Mar 15	Mar 18
Design fair and final release	Mar 19	Mar 27
Project snapshot	Sep 18	Sep 18
Project proposal	Oct 10	Oct 10
Implementation plan	Nov 7	Nov 7
Administrator design review 1	Jan 10	Jan 10
Administrator design review 2	Jan 17	Jan 17
Technical design review	Jan 30	Jan 30
Oral presentation and evaluation	Feb 9	Feb 9
Final report	Mar 20	Mar 20
Final technical release	Mar 25	Mar 25
Team showcase	Mar 27	Mar 27

Appendix B: Timeline and Documentation

https://github.com/timling234/ECE496-Capstone/blob/beta/README.md



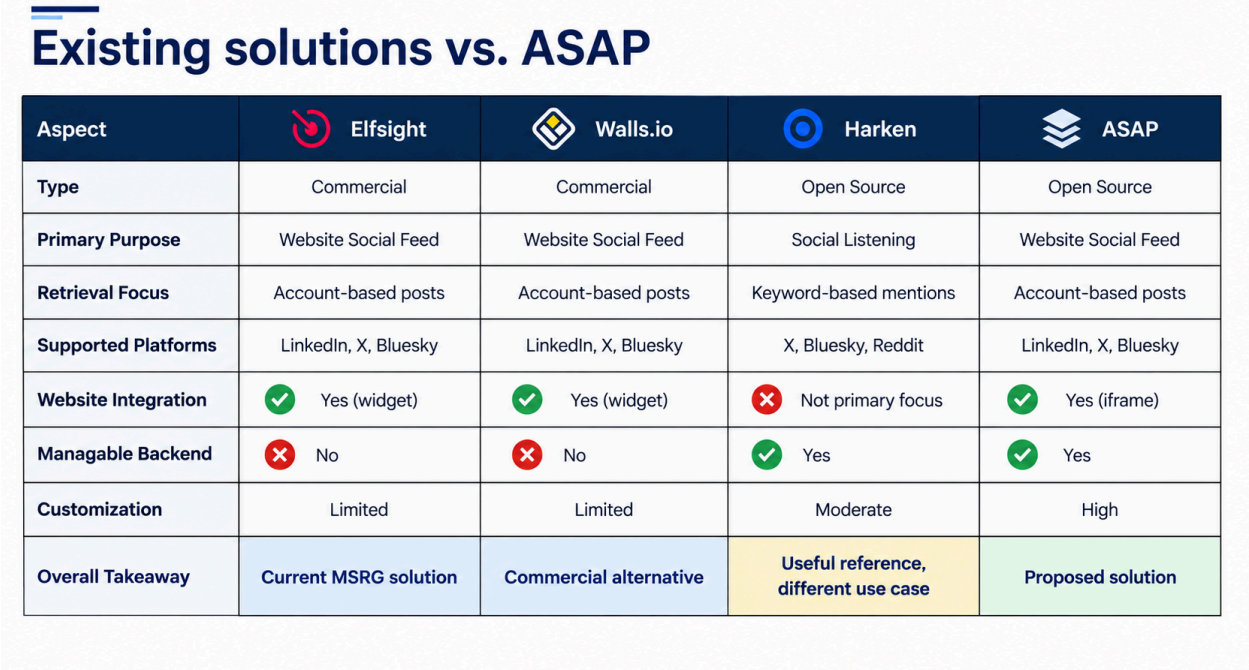
Appendix C: Preliminary Research



Initial Feasibility Check

Feasibility	Linkedin	X (Twitter)	Bluesky
API Access	Blocked	Working	Working
Content Retrieval	Blocked	Text, links & images	Text, links & images
Costs	TBD	Pay per use	Free
Overall	**Business Account Authentication**	**Able to Retrieve Data**	**Able to Retrieve Data**

Figure C.1: Initial feasibility assessment of target social-media platforms.



Existing solutions vs. ASAP

Aspect	Elfsight	Walls.io	Harken	ASAP
Type	Commercial	Commercial	Open Source	Open Source
Primary Purpose	Website Social Feed	Website Social Feed	Social Listening	Website Social Feed
Retrieval Focus	Account-based posts	Account-based posts	Keyword-based mentions	Account-based posts
Supported Platforms	LinkedIn, X, Bluesky	LinkedIn, X, Bluesky	X, Bluesky, Reddit	LinkedIn, X, Bluesky
Website Integration	Yes (widget)	Yes (widget)	Not primary focus	Yes (iframe)
Managable Backend	No	No	Yes	Yes
Customization	Limited	Limited	Moderate	High
Overall Takeaway	**Current MSRG solution**	**Commercial alternative**	**Useful reference, different use case**	**Proposed solution**

Figure C.2: Comparison of existing social-media aggregation solutions with the proposed ASAP system.

Appendix D: System Block Diagram (3 subsystems)

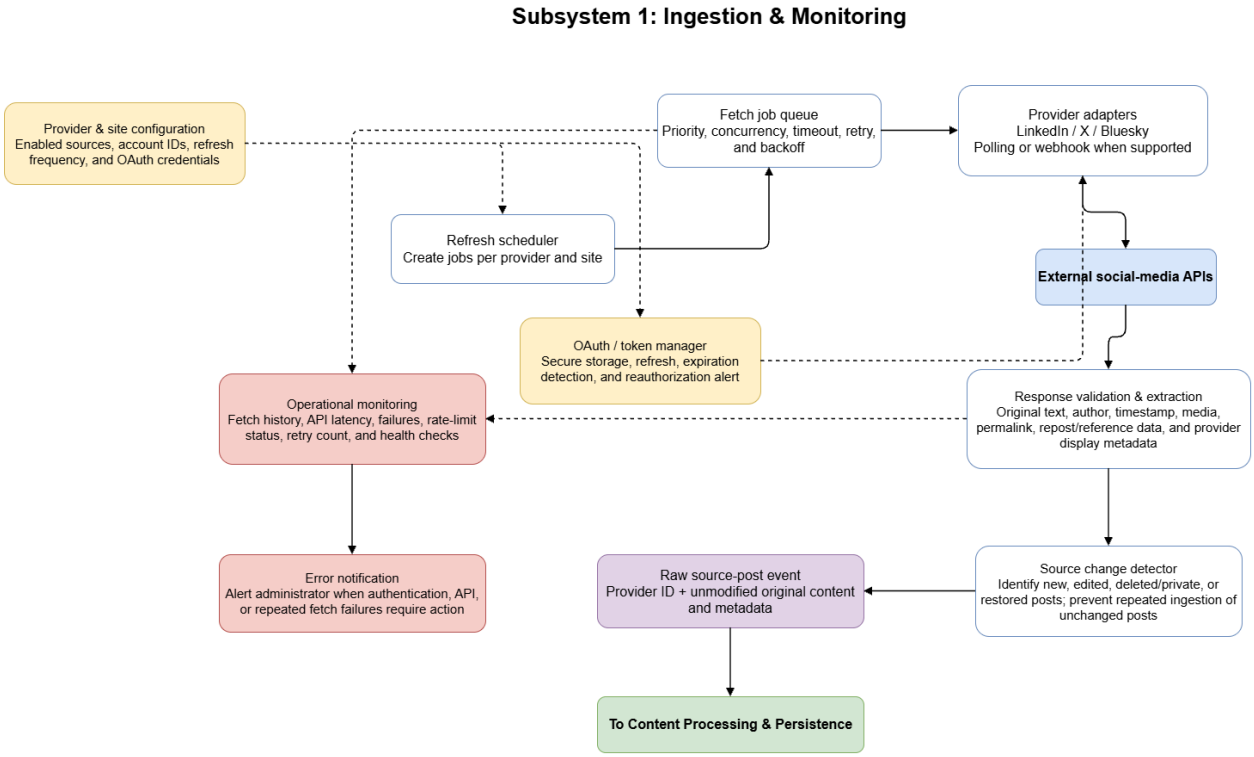


Figure D.1: Subsystem 1 : Integration & Monitoring

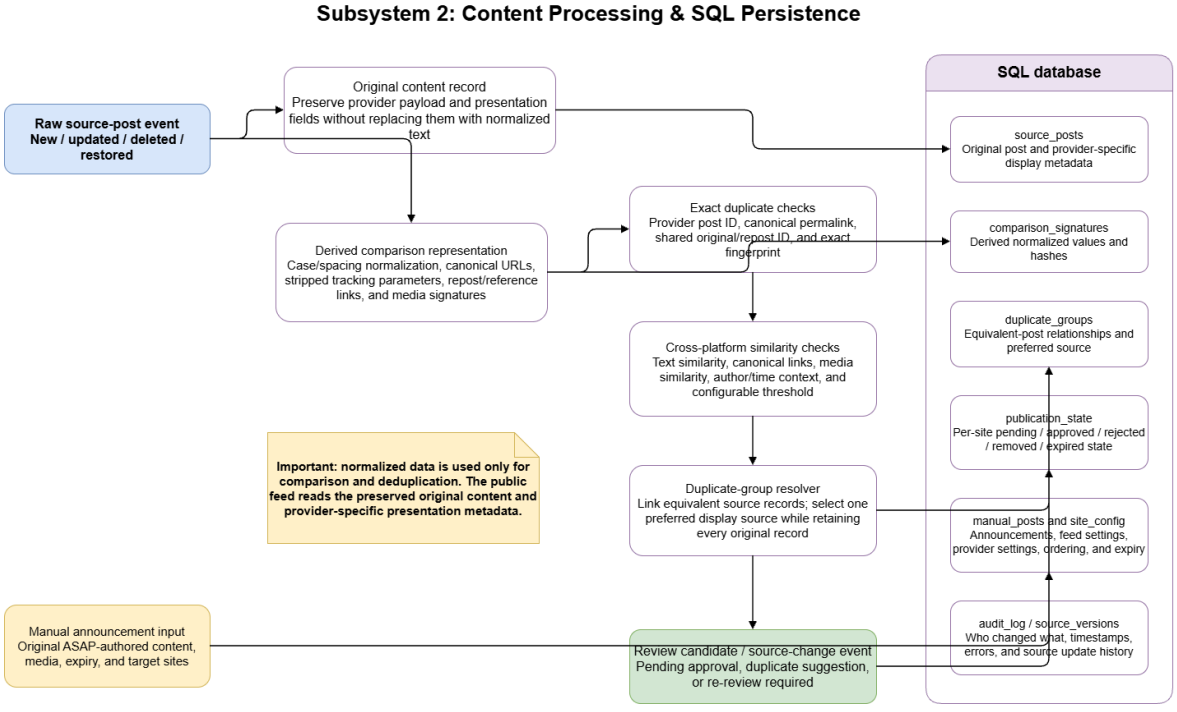


Figure D.2: Subsystem 2 : Content Processing & SQL Persistence

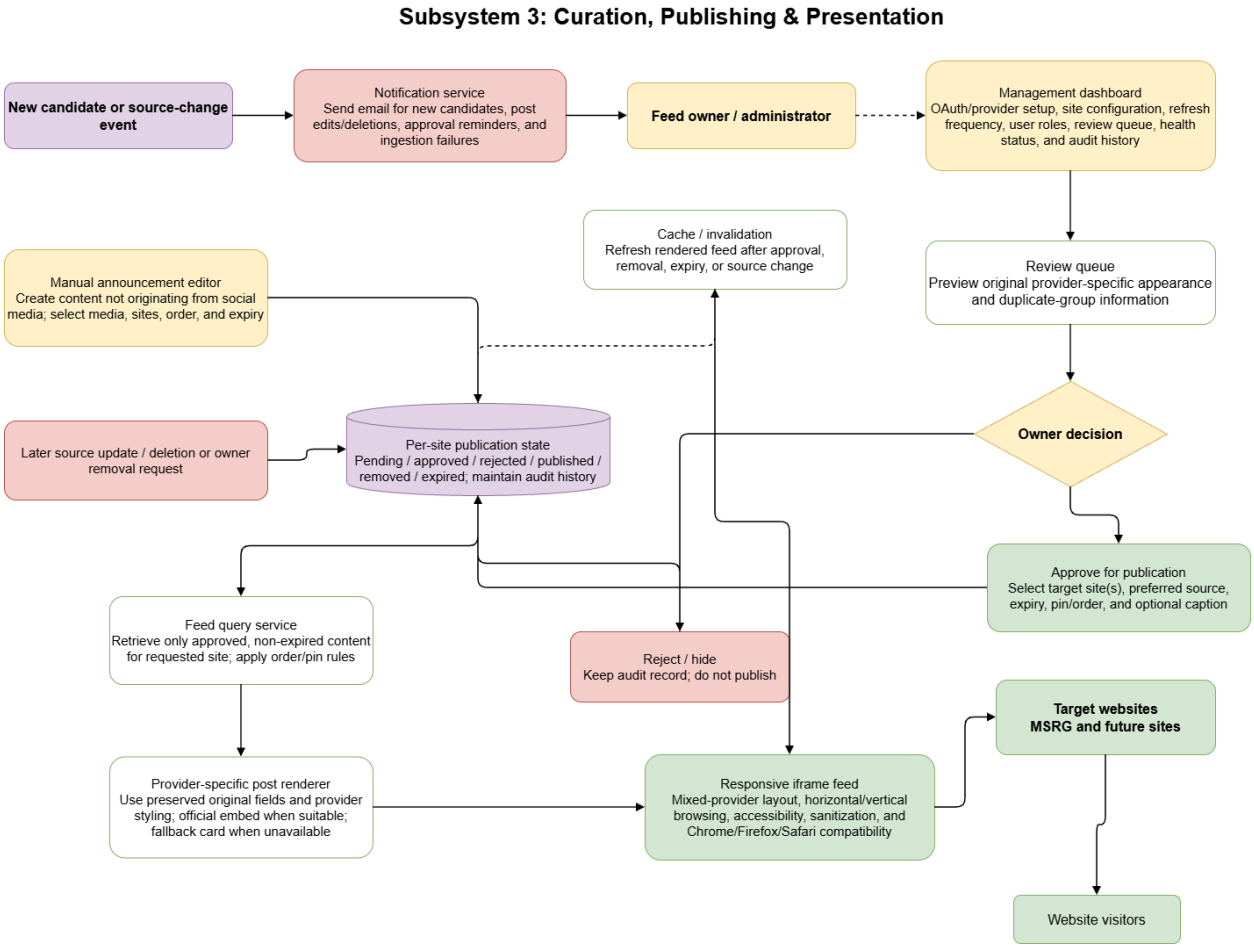


Figure D.3: Subsystem 3 : Curation, Publishing & Presentation